With the exception of some older gas cooled reactors, power reactors use ceramic pellets of UO_2, PuO_2, and ThO_2, or a mixture of these oxides, as fuels. UC has also been tested in some reactors. The size of the cylindrical pellets is $\sim 1 \times 1$ cm (diameter $\times$ height). To improve the possible burnup and to increase the temperature margin in the pin centers, the pellets are now often thinner in diameter, e.g. 0.8 cm. Fuel rods consisting of ceramic fuel pellets stacked in metallic tubes of zircaloy or stainless steel are quite temperature resistant, do not have the phase transformations of the metals, and have greater resistance to radiation effects. Unfortunately, the heat conduction is not as good as in the metallic fuel elements, and as a result rather high temperature gradients (up to $300°C\ mm^{-1}$) often exist in ceramic UO_2 elements.

21.1. PRODUCTION OF FUEL ELEMENTS

The normal raw material for production of UO_2 based fuels is enriched uranium in the form of UF_6, which is delivered in special containers. By heating the container to $\sim 100°C$ it is possible to transfer the hexafluoride as gas to the conversion plant, where UO_2 powder is produced, see Fig. 21.2. Several possible reactions can be used, e.g. hydrolysis of UF_6 by dissolution in water

$$UF_6(g) + 2H_2O \rightarrow UO_2F_2(aq) + 4HF(aq) \quad (21.1)$$

followed by precipitation with ammonia

$$2UO_2F_2(aq) + 6NH_4OH(aq) \rightarrow (NH_4)_2U_2O_7(s) + 4NH_4F(aq) + 3H_2O \quad (21.2)$$

after filtration, washing and drying the diuranate is converted to UO_2 by reduction with hydrogen at 820°C.

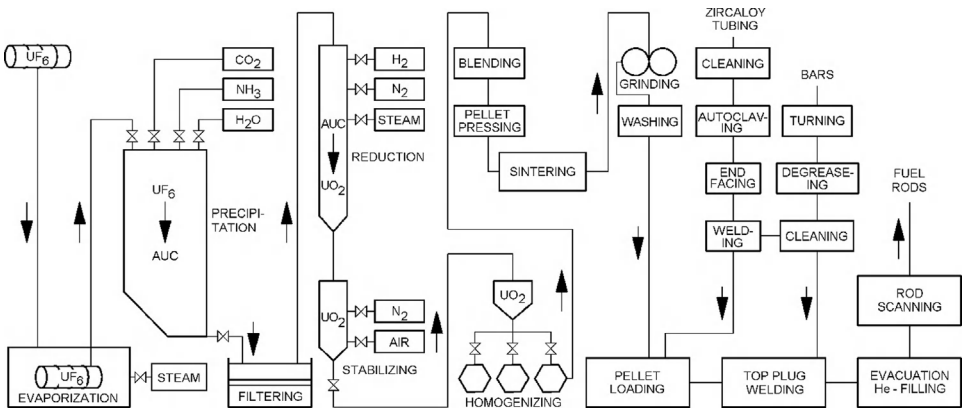


Figure 21.2 Production of UO_2-fuel rods.

Another often used reaction sequence begins with the formation and precipitation of ammonium uranyl carbonate (AUC) by reaction of UF_6 with water, CO_2 and NH_3, see Figure 21.2. The AUC is reduced to UO_2 by reduction in a fluidized bed using a mixture of hydrogen, nitrogen and steam and cooled to room temperature in a mixture of air and nitrogen.

The UO_2 powder is pressed into "green" pellets of slightly larger dimensions than the final product. These have $\sim$50% of the theoretical density. The green pellets are sintered at $\sim$1700°C in a dry H_2 atmosphere which gives a small controlled oxygen excess in the product; UO_{2+x}, where $x \sim 0.05$ is normally desired for best fuel performance. As exact dimensions of the pellets are needed in order to fit into the cladding tubes, the sintered pellets are ground to final shape.

Typically a density of $\sim$10 400 kg m^{-3} is desired, which corresponds to a porosity of $\sim$5%. If the density is too high the pellets swell excessively during irradiation due to the volume of the fission products gaseous at the operating temperature. Strongly swelling pellets may cause deformation, and failure, of the can. On the other hand a too low density can cause an initial shrinking of the pellet leading to an increased pellet – can gap, thereby reducing the heat transfer coefficient and increasing pellet temperature.

The finished pellets are put into a zircaloy tube with a welded bottom plug. Thereafter the pellet column hold-down spring and top plug are fitted and the resulting fuel pin sealed by electron beam welding, see Figures 19.12 and 21.2.

In order to improve heat transfer between pellets and can and to reduce pressure induced strain during operation, the space between pellets and tube is usually first evacuated and then pressurized with helium through a hole in the top end plug. It important to remove all traces of water as remaining water may form hydrogen by reaction with hot UO_2 during operation. The hydrogen reacts with the inside of the can forming zirconium hydride which may cause can failure.

Heat conduction between fuel and can is further improved by using *bonding materials* such as molten sodium, graphite powder, etc. The bonding material should also provide some lubrication between pellets and can. The canning material itself must not only be corrosion resistant to the coolant at all temperatures but should also not react with the fuel or the bonding material.

The can should be as thin as possible, consistent with satisfactory mechanical strength and corrosion resistance (Fig. 19.12(a)). To reduce the danger of hydride formation a protective oxide layer can be produced by autoclaving the tube before filling it with pellets. In case of UO_2 pellets in zircaloy, the bonding material, e.g. graphite, is put onto the inner surface of the zircaloy tubes before the pellets are introduced. In case of stainless steel clad fast reactor fuel the production and assembly is similar, but the bonding is usually by sodium metal.

The purpose of cladding the fuel is to protect it against corrosion and to protect the coolant from radioactive contamination by the fission products from the fuel element.

Aluminum has been used in water-cooled reactors, but at temperatures >300°C zirconium alloys show superior strength. At steam temperatures >400°C zirconium absorbs hydrogen, which increases brittleness, so stainless steel becomes preferable. In sodium cooled fast reactors, stainless steel is normally used. The most common alloys are zircaloy-2 (containing 1.58 Sn and 0.3% Cr, Ni, and Fe) and stainless steel type 302B (containing 10% Cr and 8% Ni). Stainless steel is not used at lower temperatures because of its larger neutron capture cross-section: σ 0.23 b for Al, 0.18 b for zircaloy-2, and about 3 b for 302B steel.

Metallic uranium is usually produced by conversion of UF_6 to UF_4 followed by metallothermic reduction of UF_4 by magnesium or calcium metal; however, several other methods exist. Metallic fuel is encased in a canning (cladding) of aluminum, magnesium, or their alloys. Fuel for high flux research reactors based on highly enriched uranium (>10% ^{235}U) is often made in the form of uranium metal alloys (or compounds like USi_2) canned in aluminum to improve mechanical and thermal stability.

The fuel elements for use in high temperature gas cooled reactors consist of graphite rods or balls filled with oxide or carbide kernels produced by the sol-gel process. The kernels are covered by several layers of graphite and silicon carbide achieved by pyrolyzing methane or acetylene in a fluidized bed of the kernels.

Fuel cost and performance is an important part of the economy of power reactors. Approximately 20% of the expense of the electrical production in a power reactor can be attributed to the cost of the fuel. This is due about equally to the expense of the consumption of fissile material and to the production and, when applicable, reprocessing costs or intermediate storage costs. In the fast breeder reactors, it is anticipated that fuel costs would be substantially lower because of a higher burn-up.

When mixed uranium-plutonium oxide (MOX) fuel elements are used as, for example, in plutonium recycling (<5% PuO_2) in LWRs or in fast breeders ($\leq$15% PuO_2), the UO_2–PuO_2 mixture must be very intimate. This can be achieved by coprecipitation of the tetravalent actinides, normally as oxalates, followed by calcination. However, in industrial scale production of MOX fuel a rich mixture containing 15–20% PuO_2 is first very finely ground and then diluted with coarse grained pure UO_2. The final mixture is pressed into pellets and sintered similarly to UO_2-fuel pellets. This yields a fuel which can be dissolved to 99% (or better) in nitric acid. MOX fuel elements have been regularly added to the cores of many European LWRs for many years without any technical difficulties. Mixed uranium and thorium oxide fuels have also been used in a few heavy water reactors, see §20.1.1.

Instead of ground powders, spherical fuel particles can be used as a starting material. This has advantages with respect to fabrication, reactor utilization, and fuel reprocessing. These oxide or carbide particles are very small, <1 mm in diameter. The particles are produced by the *sol–gel process*, which in principle consists of the following steps:

(i) An aqueous colloidal solution or the actinide or actinide mixture is prepared. The actinide(s) may be in the form of a hydrated complex of high concentration (3–4 M).

(ii) The solution is added to an inert solvent, which dehydrates the complex and causes the droplet to gelate. In one technique, hexamethylenetetramine, $(CH_2)_6N_4$, is added to the aqueous solution, which is added dropwise to a hot (~95°C) organic solvent. The heat causes the amine to decompose, forming NH_3, which leads to hydroxide precipitation in the droplet. The droplet dehydrates and solidifies rapidly, forming a "kernel".

(iii) The kernels are washed, air dried at 150–200°C, and – in the case of uranium – reduced by hydrogen gas at higher temperature to form UO_2.

(iv) The kernels are sintered at high temperature in an inert atmosphere.

Kernels of actinide carbides can be made in a similar manner. In many cases the kernels are covered by the addition of protective layers of graphite, silica or silicon carbide. The kernels are placed in fuel rod cans, pressed into pellets, or incorporated in a graphite matrix for use in high temperature reactors.

21.2. POWER GENERATION

During operation a large amount of heat is generated inside the fuel and has to be transferred to the coolant. Assuming a constant power per unit volume in the fuel, and fuel pins so long that longitudinal conduction can be neglected, the temperature profile can be estimated from the *specific power* (p, W m^{-3}) in the fuel from the following equations using the notation for radii from Figure 21.3.

Inside the fuel

$$T(r) = T(r_f) + p\left(r_f^2 - r^2\right)\left(4\,k_f\right)^{-1} \tag{21.3}$$

where $T(r)$ (°C) is the temperature at radius r (m) inside the fuel pellet, $T(r_f)$ (°C) the temperature at the pellet surface, r_f the pellet outer radius (m) and k_f the heat conductance in the fuel (W m^{-1} K^{-1}).

Across the gap between fuel and can

$$\Delta T_{gap} = p\,r_f\left(r_i - r_f\right)\left(2\,k_g\right)^{-1} = p\,r_f\left(2\,\alpha_g\right)^{-1} \tag{21.4}$$

where ΔT_{gap} is the temperature difference across the fuel – can gap (°C), r_i the inner radius of the can (m), k_g the heat conductivity across the gap (W m^{-1} K^{-1}) and α_g is the heat transfer coefficient across the gap (W m^{-2} K^{-1}).

For a thin can

$$\Delta T_{can} = p\,r_f^2\,(r_c - r_i)(2\,r_i\,k_c)^{-1} \tag{21.5}$$